

Envikal-Loc: Sub-10m Indoor LoRaWAN Localization with Environmental-Aware Path Loss Modeling and Adaptive RSSI Smoothing

PROBLEM

Indoor localization using LoRaWAN faces significant challenges:

- Multipath fading, dynamic obstructions (people/objects), and **environmental fluctuations** cause severe RSSI instability [1].
- Conventional models (**COST 231 Multi-Wall**) neglect environmental dynamics.
- Typical LoRaWAN localization accuracy remains between **8–20 meters** [2], which is insufficient for precision applications like hospital tracking or smart factories.

Goal: Achieve **sub-10-meter** accuracy while maintaining interpretability and efficiency.

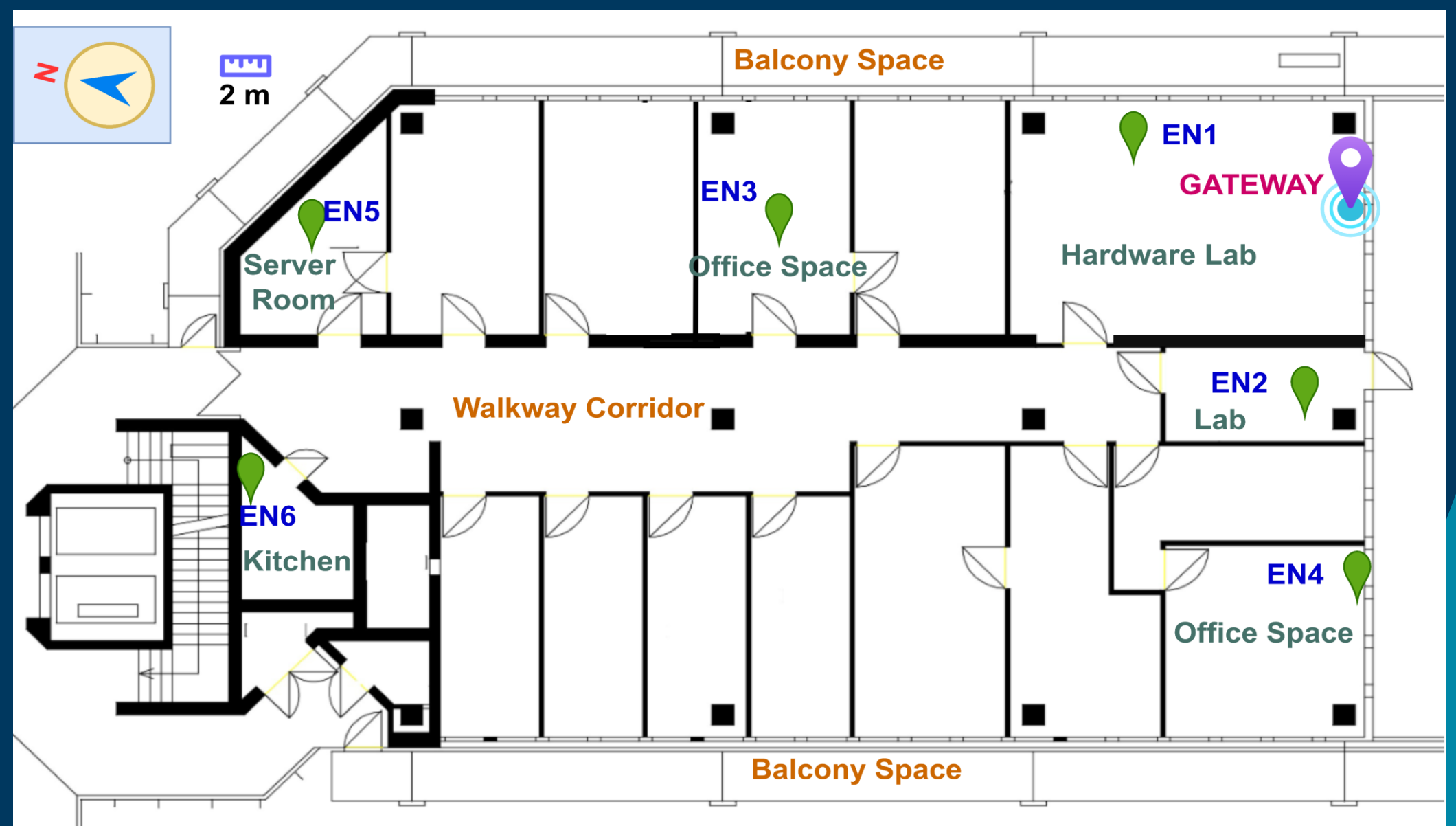


Fig. 1: Sensor Network Deployment for Data Collection (Floor Plan)

CONCEPT

We propose **Envikal-Loc**, a framework that:

- Extends traditional path-loss models by incorporating **environmental parameters** (temperature, humidity, CO₂, PM_{2.5}, pressure, SNR).
- Applies an **adaptive Kalman filter** per device to smooth RSSI fluctuations dynamically without prior knowledge of noise statistics.

This hybrid approach leverages **physical insights** and **real-time noise adaptation** for robust indoor (**Fig. 1**) LoRaWAN localization.

MODELS

COST 231 Multi-Wall Model (MWM)

- Only distance and static wall losses are considered.

Extended Model (MWM-EP)

- Adds **environmental factors** and signal-to-noise ratio (SNR).
- Multiple linear regression framework.

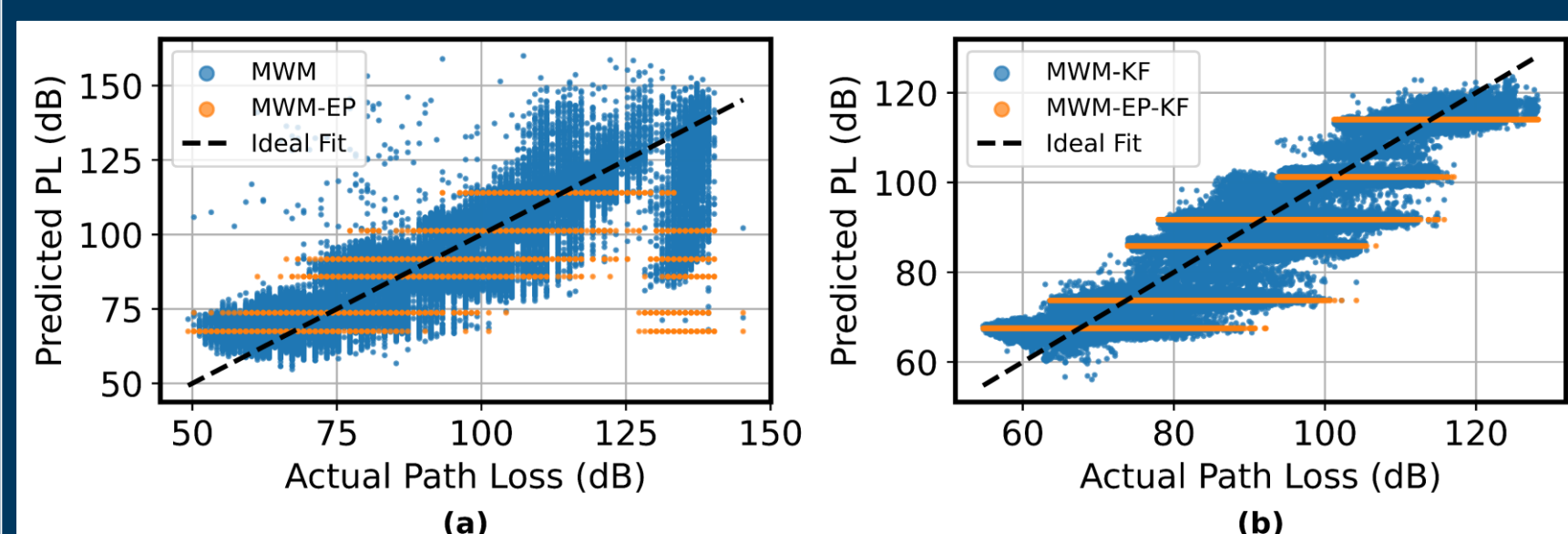
Proposed Full Model (MWM-EP-KF)

- RSSI smoothing using self-tuning **1D Kalman filter**.
- Captures persistent environmental impacts while rejecting transient noise.

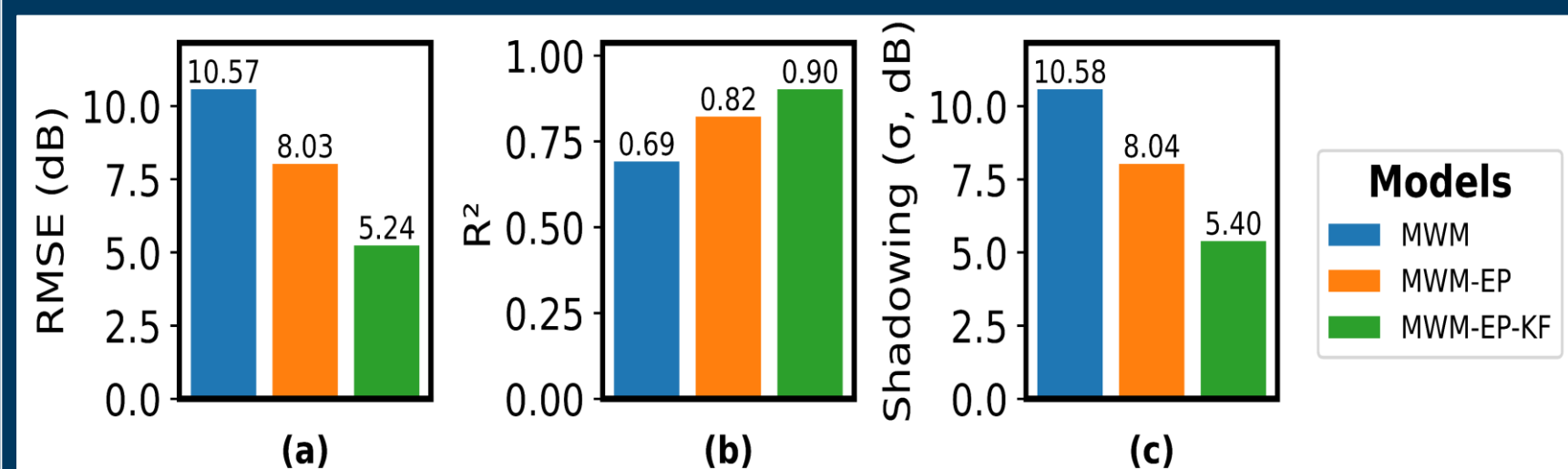
FORMULATION

$$PL = \beta + 10 \cdot n \cdot \log_{10} \left(\frac{d}{d_0} \right) + 20 \cdot \log_{10} f + \sum_{k=0}^K W_k L_k + \sum_{j=1}^J \theta_j E_j + k_s \cdot SNR + \epsilon$$

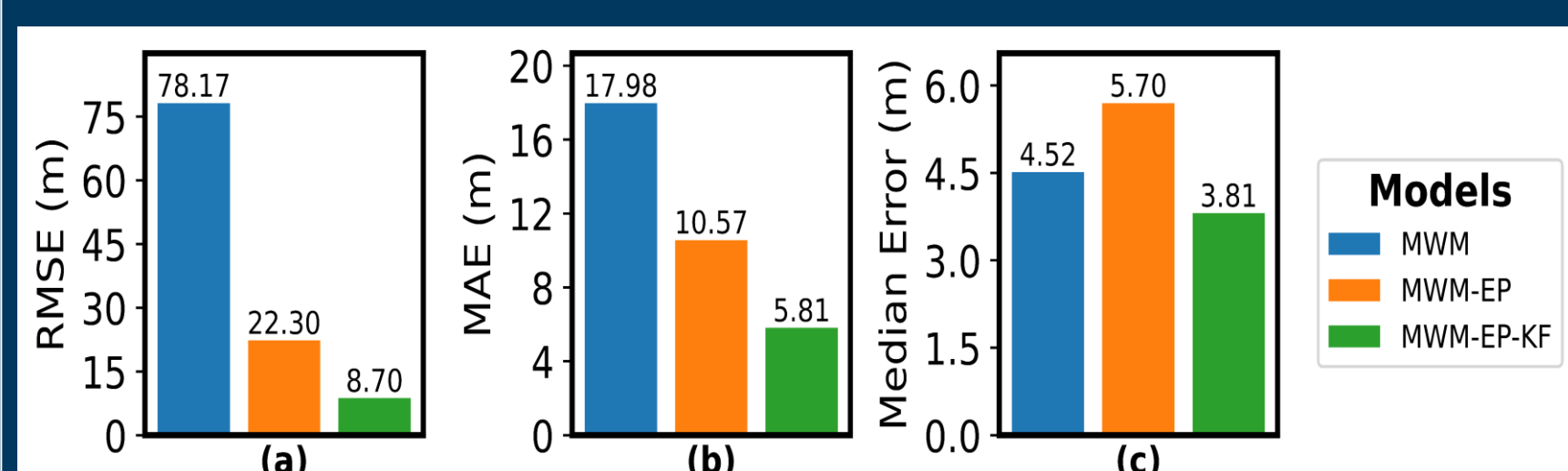
EMPIRICAL RESULTS & INSIGHTS



Comparison of predicted and actual path loss using the MWM and the MWM-EP models, with (a) the measured RSSI and (b) the filtered RSSI.



Evaluation of PLS fitting via: (a) RMSE, (b) coefficient of determination (R^2), and (c) shadowing standard deviation (σ) for the MWM, MWM-EP, and MWM-EP-KF models.



Comparison of distance estimation error across MWM, MWM-EP, and MWM-EP-KF models using: (a) RMSE, (b) MAE, and (c) Median Error.

IMPACT AND FUTURE WORK

- Envikal-Loc enhances indoor LoRaWAN localization by modeling **environmental** dynamics, **adaptively** filtering signal noise, and delivering accurate, **interpretable** results.

Future IoT Tracking Applications:

- Hospitals,
- Smart factories,
- Logistics.

Future Work:

- Multi-gateway support
- Mobile nodes
- Sensor fusion.

REFERENCES

- <https://doi.org/10.3390/enq4020085>
- <https://doi.org/10.1109/ACCESS.2024.3509516>